

Policy Brief

November 13, 2024

Saving Children: Reducing Phthalates in Children's Toys

Executive Summary:

To the Chairman and Commissioners of the Consumer Product Safety Commission and the Administrator of the Environmental Protection Agency:

Phthalates are a type of plasticizer used to make plastics soft, more flexible, and durable, and they are commonly found in numerous consumer products including, cosmetic products, food packaging, and children's toys like teething items and rubber ducks. Phthalates have been used for decades, but scientists have discovered that this chemical has detrimental effects on human health. It has been shown to be an endocrine disruptor, with long-lasting impacts, especially for children, who are often exposed to phthalates in toys that they frequently mouth. This exposure can significantly affect the health and development in children.¹ Due to these potential health impacts, phthalate regulation has emerged as a critical public health issue. At the federal level, the Consumer Product Safety Improvement Act (CPSIA) has prohibited the manufacture, sale, and distribution of children's products that contain more than 0.1% of phthalates. This act includes the following phthalates: DEHP, DBP, BBP, DINP, DIDP, and DnOP.² In response, states have adopted different policies limiting or banning phthalates. This has prompted the need to increase phthalate regulation through certification under the CPSC, regular monitoring and reporting of industry phthalate levels, increase education surrounding the issue, and mandate fines for violations. This policy analysis examines the effectiveness and scope of increased federal-level phthalate regulations to determine what the best policy adaptation is to prevent exposure to phthalates in children.

Statement of Problem & Background:

Phthalates, a group of chemicals, have been incorporated into a variety of products for the past century. It was used primarily to strengthen and increase the durability of plastics as well as to retain scent and increase skin penetration in perfumes and lotions, phthalates were commonly incorporated into cookware, cosmetics, toys, and packaging.^{2,3} In the late 1900s, research began to demonstrate that phthalates can mimic human hormones, consequently altering very delicate balances in the human body and labeling them as "endocrine-disrupting chemicals".⁴ Additionally, exposure to phthalates has been shown to lead to severe developmental issues, notably impairment of brain development, changes in fertility, genital defects, respiratory problems, obesity, and diabetes.^{4,5} Phthalates, present in concentrations up to 44%, were found to be present in children's toys and care products such as teethingers, via which phthalates were easily digested by young children.⁶ Research also shows the role of phthalates in increasing the risk of allergic diseases and reactions amongst children with high exposure, causing inflammation within their airways.⁷ The behaviors of children have influenced their increase in exposure to these chemicals, and because children have a larger surface area in comparison to body mass with adults, their vulnerability to ingestion and skin exposures increases their exposure as well.⁷

Phthalates can enter a child's body through ingestion and inhalation making them susceptible to exposure from common household toys and food they may consume. Phthalates are not chemically bound to the products they're used in and can easily contaminate the surrounding environment. Early and constant exposure can have multiple effects over time, including long-term developmental issues due to the disruptive impact of phthalates on the endocrine system's hormone regulation. This can impact

cognitive and behavioral outcomes as well as developmental impairments.⁸ Research indicates that prenatal exposure to phthalates can affect brain regions such as emotion regulation and attention. This hormonal disruption during development can lead to observable behavioral changes (i.e. increased impulsivity, difficulty focusing, and challenges with social interaction), as chemical exposure affects dopamine and thyroid regulation in the hippocampus.⁹ These findings highlight the need for stricter regulation and awareness to reduce phthalate exposure, particularly in vulnerable populations like young children.

After growing research supporting the link between phthalate exposure and negative health effects, the US banned the manufacture and distribution of children's toys and childcare products containing over 1000ppm or 0.1% of phthalates including DEHP, DBP, BBP, DIBP, DPENP, HEXP, and DCHP with the *Consumer Product Safety Improvement Act* (CPSIA) of 2008.^{1,2} The act also requires testing of products to prove the concentration of phthalates, a certificate indicating compliance with the regulations, and identifying information to allow for tracking of the product.² Prior to the passing of the CPSIA, phthalates were not federally regulated. California, New York, Maine, Oregon, Washington, and Vermont have adopted their own state-level phthalate regulations.^{2,10} Signed into law in 1986, California's Proposition 65 requires the state to keep a list of carcinogenic chemicals found day-to-day and update it annually at a minimum. Additionally, Proposition 65 calls for businesses to provide warning labels on products and online product display pages detailing the risks of chemical exposure.^{11,12} Vermont similarly passed the *Chemicals of High Concern to Children* (CHCC) Law in 2017, requiring manufacturers using chemicals labeled as dangerous to children by the State of Vermont to report to the State Health Department.¹³

Despite federal regulations on phthalates, most states have not taken additional measures to limit phthalate exposure. In Texas for example, the Texas Association of Manufacturers, along with others, formed a petition requesting the U.S. Court of Appeals for the Fifth Circuit to review the CPSIA final rule on phthalates.¹⁴ In March of 2021, the court sent the rule back to the Consumer Product Safety Commission (CPSC) and asked the agency to fix two issues that it identified within the case of *Tex. Ass'n of Mfrs. v. CPSC*, 989 F.3d 368 (5th Cir. 2021).¹⁴ The court found that in this case, the CPSIA final rule did not: (1) provide enough notice and comment when it changed its main justification from the proposed rule to the final version; and (2) consider the costs and benefits of making the interim ban on diisononyl phthalate (DINP) permanent.¹⁴ The court also did not cancel the final rule, thus leaving it in effect since 2018.¹⁴ As a result, the flaws within the regulation of DINP in Texas have allowed for a potential of increased exposure for children when it comes to phthalates, something that The American Chemistry Council (ACC) and a group made up of the Natural Resources Defense Council, the Environmental Justice Health Alliance for Chemistry Policy Reform as well as other advocates condemned.¹⁴

Policy Alternatives:

The current policy landscape has helped limit the exposure of phthalates in children since 2008, however, in 2024 there could be improvements to fully reduce the risk of exposure and hold companies accountable, and strengthen existing regulations. These policies will be analyzed based on affordability, effectiveness, and political feasibility. For affordability, how expensive would it be to implement these state policies regarding phthalates and children's health? For effectiveness, how will these policies successfully improve children's health outcomes in the U.S.? For political feasibility, will the policy be successful in becoming a law and addressing the issue of chemical exposure to phthalates? See Figure 1: Comparison of Policy Alternatives.

Policy 1: Senate Bill 2120 (S.2120)

Senate Bill 2120 was introduced in the 113th Congress on March 12, 2014. This bill aims to strengthen the regulations of phthalates in children's products in order to reduce the exposure of phthalates in children.¹⁵ This bill was proposed to add an amendment to the CPSIA by changing the

language from “children’s toy or child care article” to “children’s product” in order to expand the scope of exposure from children’s toys to all kinds of children’s products.¹⁵ It also wanted to change the language from “or benzyl butyl phthalate (BBP)” to “benzyl butyl phthalate (BBP), diisononyl phthalate (DINP), diisodecyl phthalate (DIDP), or di-n-octyl phthalate (DnOP)” in order to strengthen the ban on additional phthalates.¹⁵ By adding these amendments to the CPSIA it would reduce the risk of exposure of phthalates in children and improve the current ban that is in place. Passing S.2120 would increase the amount of children’s products being recalled due to these new products being under investigation and new phthalates being on the banned list. This would increase costs for businesses since they would have to recall and reformulate products in order to stay compliant, ranking this policy low on the affordability scale. S.2120 would contribute to a reduction in exposure because of the expansion of phthalates under the CPSIA which ranks this policy high on effectiveness. This policy ranks low on political feasibility because the Consumer Product Safety Commission previously came to the decision that DIDP and DNOP did not show a significant level of health risks.¹⁶

Policy 2: New York Assembly Bill A432 (A432)

New York’s legislation introduced A432 during the 2023-2024 legislative session which targets phthalates in children’s products, specifically products that could be mouthed by children under three years of age. A432 aims to ban the manufacture, distribution, and sale of toys and other childcare products that contain phthalates (e.g., DEHP, DBP, BBP, DINP, DIDP, and DnOP) in concentrations above 0.1%, similar to CPSIA.¹⁷ In January 2024, A432 was referred to the Consumer Affairs and Protection Committee where the bill remains active. A432 differs from the CPSIA in the scope of chemicals regulated and the enforcement and penalties for non-compliance. A432 requires manufacturers to use the “least toxic alternative” to phthalates and prohibits substitutes that are carcinogenic or reproductive toxicants, which is not specified in the CPSIA.^{11,17} The CPSIA does not impose fines for non-compliance compared to A432 where civil penalties cost \$10,000 per day.^{11,17} The CPSIA and A432 align greatly in prohibiting phthalates in children’s products but A432 extends protections with specific mandates and stricter penalties for non-compliance. Implementing A432 would increase manufacturing costs due to having to avoid using phthalates and other harmful substitutes. Civil penalties of \$10,000 per day may pose a high financial risk for small businesses and manufacturers, making A432 low on the affordability criteria scale. A432 prohibiting the usage of six types of phthalates, carcinogenic substitutes, and reproductive toxicants would result in positive health outcomes for children, making A432 high on the effectiveness criteria scale. Other strict policies like A432 have been implemented in states like California and Vermont, highlighting the political feasibility of its implementation.¹⁸ However, opposition from industry stakeholders may block the passing of A432, making A432 medium on the political feasibility criteria scale.

Policy 3: Children’s Chemical Risk Reduction Act, H.R. 4030

The Children’s Chemical Risk Reduction Act, H.R. 4030 introduced by U.S. Representative Darlene Hooley in 2007 during the 110th Congress would prohibit the manufacturing, selling, and distributing of children’s products and child care articles containing any phthalates, ultimately reducing children’s exposure to phthalates.¹⁹ The bill requires the U.S. to ban these products and treat them as hazardous substances under the Federal Hazardous Substances Act (FSHA). The FSHA allows the CPSC to ban hazardous products such as children’s toys containing phthalates.^{16,20} The act also prohibits the manufacturers from using harmful alternatives to phthalates.¹⁹ This act differs from the CPSIA as it completely bans the use of phthalates rather than setting limits of less than 0.1% and directly targets phthalates in children’s products rather than a large group of toxic chemicals.^{11,19} Implementation of the Children’s Chemical Risk Reduction Act will increase manufacturing costs as it forces children’s product and child care article manufacturers to buy and use nonhazardous materials receiving a low rating for affordability criteria. This act is highly efficient at preventing phthalate exposure to children by banning the use of all phthalates in children’s products treated under the Federal Hazardous Substances Act. In

terms of political longevity, the bill's death after referral to the House Committee of Energy and Commerce leads to a low rating on the criteria scale. Furthermore, the introduction of the bill by Democratic Representative Darlene Hooley posed disagreements by opposing political parties and industry stakeholders who produce children's products with these materials.¹⁹

Figure 1: Comparison of Policy Alternatives

	Policy 1 - Senate Bill 2120: Aims to amend CPSIA by changing the ban to all children's products and expanding the ban to additional phthalates.	Policy 2 - New York's AB 432: aims to prohibit concentrations of phthalates above 0.1% and imposes strict mandates and penalties for non-compliance.	Policy 3 - Children's Chemical Risk Reduction Act H.R. 4030: aims to fully ban the manufacturing, selling, and distribution of children's products.
Affordability	Low	Low	Low
Effectiveness	High	High	High
Political Feasibility	Low	Medium	Low

Policy Recommendations:

After examining the current policies, we recommend strengthening the Consumer Product Safety Improvement Act (CPSIA) as well as introducing comprehensive guidelines that expand on the provisions of New York's Assembly Bill A432, Senate Bill 2120, and Children's Chemical Risk Reduction Act H.R. 4030. The following components are essential for successful implementation based on the most beneficial aspects of each bill.

- 1. Federal legislation amendments:** CPSIA should be amended by Congress to expand the prohibition of phthalates- including the additional phthalates (i.e. di-isononyl phthalate, di-iso- 7 decyl phthalate or di-n-octyl phthalate) listed in New York's A432 and Senate Bill 2120.^{15,17} In addition, we recommend adding BBzP to the list as research has found that it can increase the likelihood of having asthma or other respiratory illnesses.⁶ This will ban additional concentrations above 0.1% in children's products or items that come into contact with children's routes of exposure.
- 2. Education and outreach:** Currently, there is minimal population knowledge surrounding phthalates, especially in children's products. The EPA and CPSC should implement a nationwide campaign through TV segments, facilitating town halls, and in schools to educate consumers, retailers, and manufacturers about the health risks of phthalates, why compliance is important, and the availability of safer alternatives.
- 3. Regular monitoring and reporting:** The CPSC must mandate regular certification every year and yearly testing for phthalate levels in children's products. This will ensure ongoing compliance and track exposure reductions over time as there is minimal data collection.
- 4. Non-compliance mandates:** As New York A432 enforces non-compliance fees of up to \$10,000 per day, the recommended policy advocates for civil penalties of \$15,000 per day to deter companies from violating the policy. This will incentivize companies to switch to phthalate-free products or remain under the 0.1% level for the safety of children in decreasing the risk of endocrine disruption and development issues, similar to the model of the Children's Chemical Risk Reduction Act H.R. 4030 for banning phthalate concentrations.²¹

Potential obstacles and strategies to overcome them include:

1. **Industry pushback:** Increased costs and production changes can lead to resistance from toy manufacturers. This is needed as old toys can contain exceeding amounts that are against legal levels, such as over 80% of old toys in a study containing high amounts of toxic chemicals and phthalates.²² To counter this, initial financial support via tax credits and federal grants during the period of transition for 6 months can mitigate these effects. This will incentivize companies to transition to less phthalate concentrations or phthalate-free alternatives while following regulations.²³
2. **Enforcement challenges:** Due to variable resources and regulatory priorities across states, consistent enforcement will be difficult. To alleviate this issue, a centralized federal system that monitors compliance tracking and penalties will standardize enforcement under the CPSC. This will allow for federal resources to support state-level enforcement efforts through centralized regulation and provide comprehensive resources for alternatives.²³

In order to eliminate harmful phthalate exposure nationwide, phthalate restrictions must apply to a broader range of children's products to reduce toxicity and hormone disruption in children as well as in aquatic life.²⁴ The criteria of medium cost, high effectiveness, and medium political feasibility highlight the necessary components that prioritize the safety of children. Although there will be initial high costs to offset the transition period, it will be within the toy companies' duties to fulfill new requirements by undergoing regular phthalate testing and certification by the CPSC with the expansion of products under the ban (including toys or children's products) or receive a fine of up to \$15,000 per day for violations. The financial costs and policy control are essential for creating safer toys under the increased regulation and educational efforts to attain improved environmental and biological health for children at a national level. The change towards new policies similarly modeled to state-wide advocacy and supervision under federal law will result in better health outcomes. Therefore, the increased accountability and viable measures of implementation will improve health for children with a lowered risk for cancer, respiratory illness, endocrine disruption, and neurodevelopment within toys.^{4, 5, 25}

References:

1. Wang Y, Qian H. Phthalates and their impacts on human health. *Healthcare*. 2021;9(5):603. doi:10.3390/healthcare9050603
2. Office of the Commissioner. Phthalates in cosmetics. U.S. Food And Drug Administration. Published May 19, 2022.
3. What are phthalates? Environmental Working Group. Published September 24, 2024. <https://www.ewg.org/news-insights/news/2024/09/what-are-phthalates>
4. Fighting phthalates. Published June 10, 2024. <https://www.nrdc.org/stories/fighting-phthalates>
5. Kim DY, Chun SH, Jung Y, et al. Phthalate plasticizers in children's products and estimation of exposure: Importance of migration rate. *International Journal of Environmental Research and Public Health*. 2020;17(22):8582. doi:10.3390/ijerph17228582
6. Braun JM, Sathyanarayana S, Hauser R. Phthalate exposure and children's health. *Current Opinion in Pediatrics*. 2013;25(2):247-254. doi:10.1097/mop.0b013e32835e1eb6

7. Bloom MS, Clark JM, Pearce JL, et al. Impact of Skin Care Products on Phthalates and Phthalate Replacements in Children: the ECHO-FGS. *Environmental Health Perspectives*. 2024;132(9). doi:10.1289/ehp13937
8. Martin L, Zhang Y, First O, et al. Lifestyle interventions to reduce endocrine-disrupting phthalate and phenol exposures among reproductive age men and women: A review and future steps. *Environment International*. 2022;170:107576. doi:10.1016/j.envint.2022.107576
9. Daniel S, Balalian AA, Insel BJ, et al. Prenatal and early childhood exposure to phthalates and childhood behavior at age 7 years. *Environment International*. 2020;143:105894. doi:10.1016/j.envint.2020.105894
10. Phthalates Impair Children's Brain Development, Scientists Call for. Columbia University Mailman School of Public Health. Published October 3, 2022. <https://www.publichealth.columbia.edu/news/phthalates-impair-childrens-brain-development-scientists-call-immediate-action>
11. The Consumer Product Safety Improvement Act (CPSIA). U.S. Consumer Product Safety Commission. <https://www.cpsc.gov/Regulations-Laws--Standards/Statutes/The-Consumer-Product-Safety-Improvement-Act>
12. Proposition 65 Warnings Website - Your right to know. Proposition 65 Warnings Website. <https://www.p65warnings.ca.gov/>
13. Chemical Disclosure Program for children's products | Vermont Department of Health. Published October 24, 2024.
14. Prohibition of children's toys and child care articles containing specified phthalates. Federal Register. Published December 5, 2022. <https://www.federalregister.gov/documents/2022/12/05/2022-25811/prohibition-of-childrens-toys-and-child-care-articles-containing-specified-phthalates>
15. S.2120 - 113th Congress (2013-2014): A bill to expand the prohibition on the manufacture, distribution, and importation of children's products that contain phthalates, and for other purposes. Congress.gov. Published 2014. Accessed November 8, 2024. <https://www.congress.gov/bill/113th-congress/senate-bill/2120/text>
16. CPSC Prohibits Certain Phthalates in Children's Toys and Child Care Products. U.S. Consumer Product Safety Commission. Published October 20, 2017. Accessed November 9, 2024. <https://www.cpsc.gov/Newsroom/News-Releases/2018/CPSC-Prohibits-Certain-Phthalates-in-Childrens-Toys-and-Child-Care-Products#:~:text=The%20CPSIA%20also%20established%20a,children%20with%20sucking%20or%20teething>
17. New York Assembly Bill A432 (2023-2024): Prohibits the manufacture, distribution and sale of toys and childcare products containing phthalates <https://www.nysenate.gov/legislation/bills/2023/A432>.
18. Shen Y. Phthalate regulations in the United States: An overview. Compliance Gate. May 19, 2023. <https://www.compliancegate.com/phthalate-regulations-united-states/>.
19. Hooley D. *Children's Chemical Risk Reduction Act.*; 2007. Accessed November 3, 2024. <https://www.congress.gov/bill/110th-congress/house-bill/4030/titles>
20. Statutes. CPSC.gov. Published January 9, 2016. <https://www.cpsc.gov/Regulations-Laws--Standards/Statutes>
21. Boissiere-O'Neill T, Lee WR, Blake TL, Sly PD, Dwan Vilcins. Exposure to endocrine-disrupting plasticizers and lung function in children and adolescents: A systematic review and meta-analysis. *Environmental research*. 2024;243:117751-117751. doi:<https://doi.org/10.1016/j.envres.2023.117751>
22. Carney Almroth B, Slunge D. Circular economy could expose children to hazardous phthalates and chlorinated paraffins via old toys and childcare articles. *Journal of Hazardous Materials Advances*. 2022;7:100107. doi:<https://doi.org/10.1016/j.hazadv.2022.100107>

23. University of Massachusetts Lowell. Phthalates and Their Alternatives: Health and Environmental Concerns. uml. Published January 2011. Accessed November 13, 2024.
https://www.uml.edu/docs/Phthalate%20and%20their%20Alternatives_tcm18-229903.pdf
24. Phthalates - Washington State Department of Ecology. ecology.wa.gov.
<https://ecology.wa.gov/Waste-Toxics/Reducing-toxic-chemicals/Addressing-priority-toxic-chemicals/Phthalates>
25. Kirchnawy C, Hager F, Osorio Piniella V, et al. Potential endocrine disrupting properties of toys for babies and infants. *PLoS One*. 2020;15(4):e0231171. Published 2020 Apr 3.
doi:10.1371/journal.pone.0231171
26. Substitutes for phthalates in toys: no health risk detected for children under three years of age. Anses - Agence nationale de sécurité sanitaire de l'alimentation, de l'environnement et du travail. Published October 17, 2016. Accessed November 14, 2024.
<https://www.anses.fr/en/content/substitutes-phthalates-toys-no-health-risk-detected-children-under-three-years-age-0>

Group Contributions:

The group evenly distributed the work with Sandy working on the Executive Summary section. This was followed by Geselle, Priyanka, and Natalia working on the Statement of the Problem and Background section. Juliette, Jackie, and Jacob each contributed to a policy alternative by addressing the confront tradeoffs and various perspectives. The paper concluded with Manjari and Michelle writing the policy recommendations section. All members contributed to proofreading, citing, making final edits, and design aspects.